Introduction

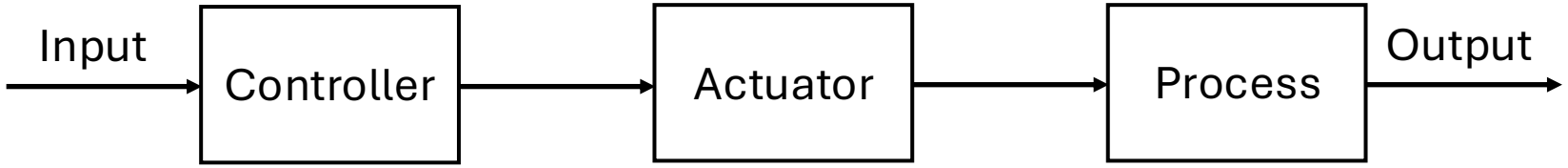
Control Engineering

What Is Control Engineering?

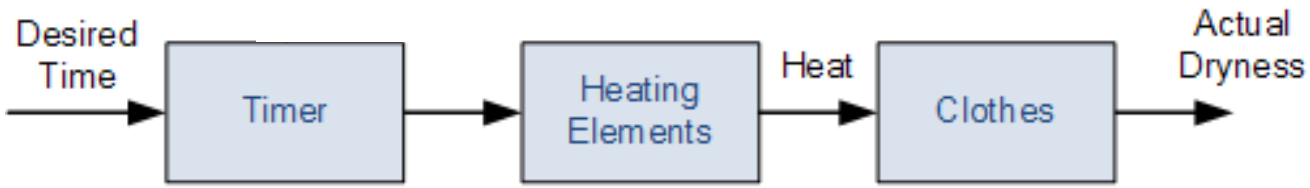
Making processes behave as desired

- Objective: regulate, track, and stabilize dynamic processes
- Control is about **decision-making over time**

Open-Loop Control



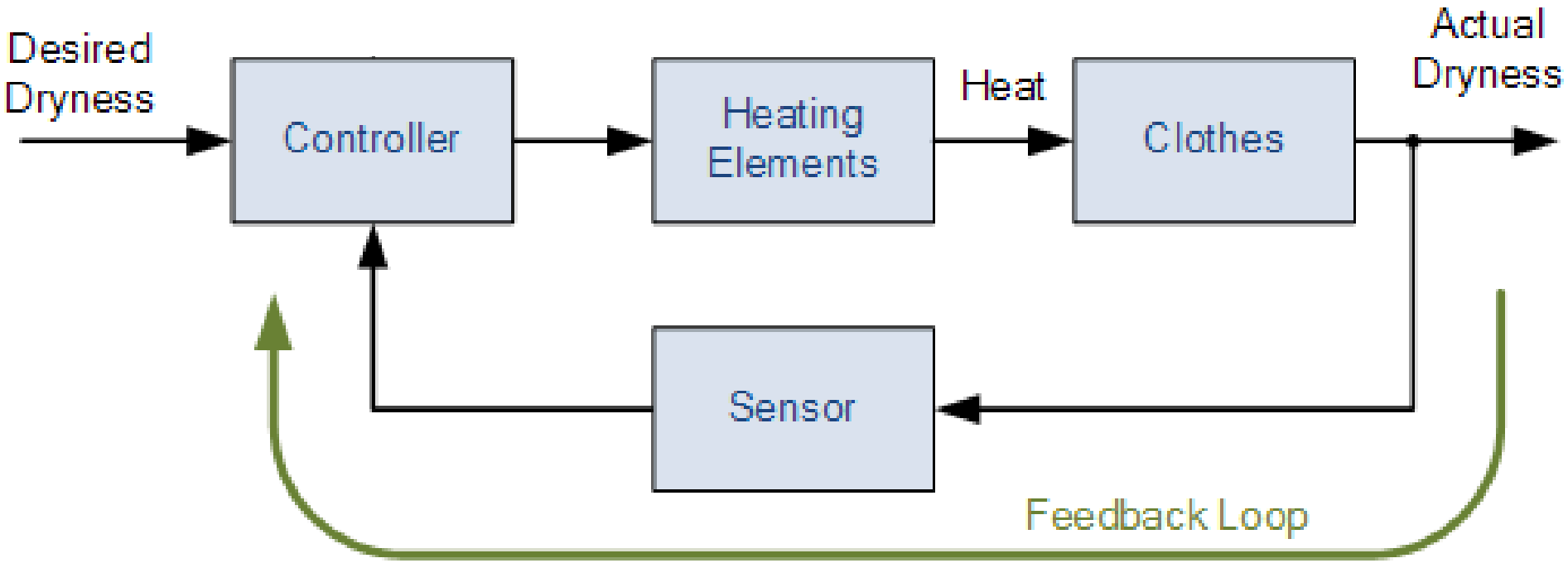
Control action does **not depend on output**



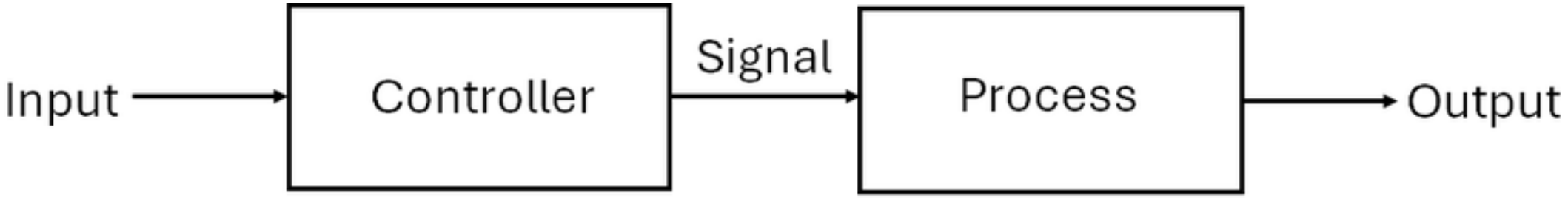
programmed to turn on at set times

- No feedback
- Simple, cheap, but inaccurate
- Cannot correct disturbances or model errors

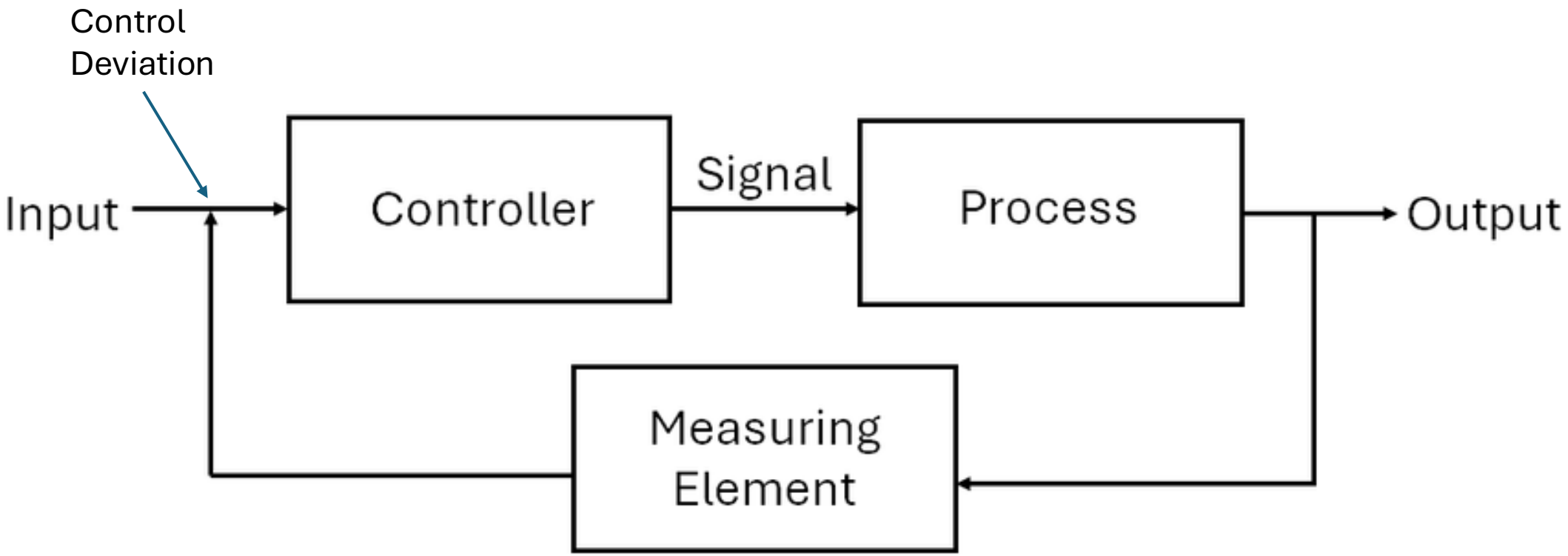
Closed-Loop Control



Output is measured and fed back



Open-Loop Control System

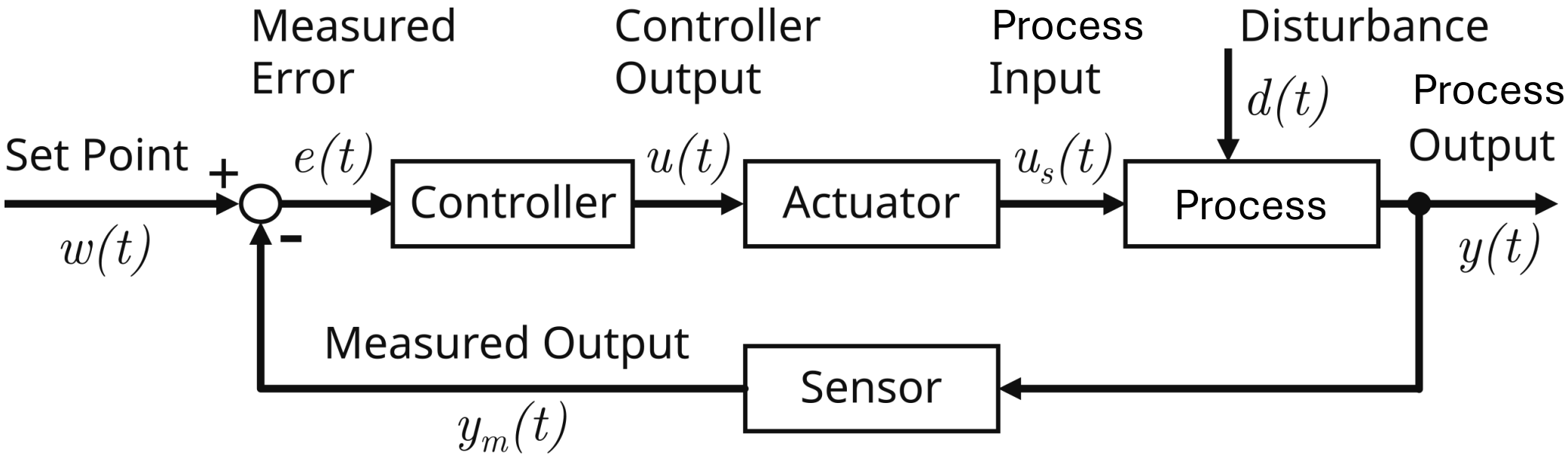


Closed-Loop Control System

Closed-Loop Control

- Controller compares reference and output
- Error signal drives control action

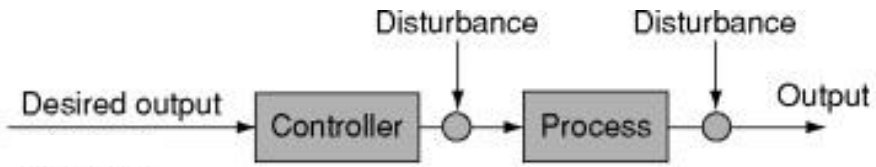
maintains temperature near setpoint

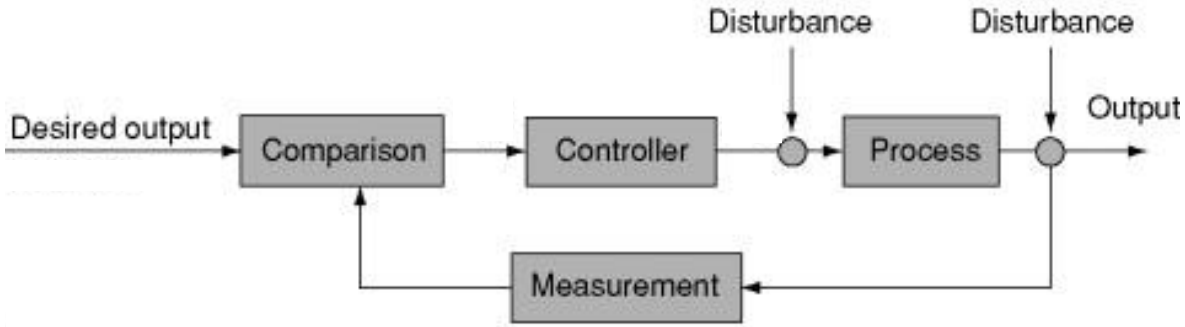


→ Improves accuracy and robustness

Sensors and Actuators

- Sensors measure system output (position, speed, temperature)
- Actuators apply control input (motor torque, valve opening)
- Measurement noise and delays are unavoidable
- Control design must account for imperfections





Feedback: Why It Works

- Reduces sensitivity to disturbances
- Reduces effect of model uncertainty
- Improves tracking accuracy
- But can also destabilize systems if poorly designed

Control Engineering Applications

- Robotics: motion control, balance, manipulation
- Automation: process control, manufacturing lines
- Automotive: cruise control, ABS, engine control
- Energy systems, aerospace, biomedical engineering
- …

Typical Control Engineering Workflow

- Understand the physical process
- Build a mathematical model
- Design a controller (regarding the process behavior)
- Simulate and analyze performance
- Implement and test

Software Tools for This Course

Python ecosystem:

- numpy, scipy
- matplotlib
- python-control

Advantages: open-source, reproducible, widely used

Course Structure and Goals

- Understand core control concepts
- Analyze system behavior
- Design classical and modern controllers
- Bridge theory and implementation

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods